Highway Litter Sampling

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Placeholder slide

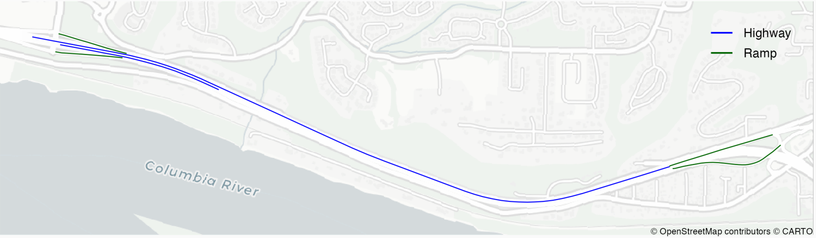
I’ll add some slides when they’re done – Addison

Sampling Design

Approach:

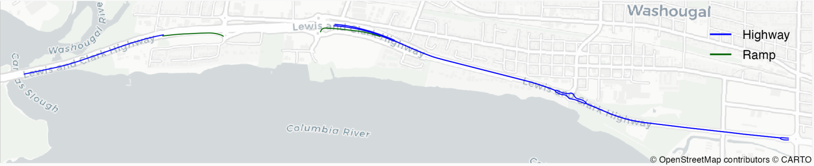
- Determine areas (strata) to use to group all the highways and ramps
- Assumption: trash in a given area is relatively similar anywhere in the area, but possibly different from other areas
- Pick some number of segments from each area to use in the sample

West Side Areas



- 2 highway stretches (north/westbound, south/eastbound)
- 2 pairs of ramps

East Side Areas



- 3 highway stretches
- 1 pair of ramps

Groupings

Approach:

- 5 highway areas and 5 samples: take 1 sample from each
- 3 ramp pairs and 3 samples: take 1 sample from each pair
- Make sure at least 1 on-ramp and 1 off-ramp included

Sampling Execution

Idea

Do sampling entirely in software using geospatial data from WSDOT

We use data from the **WSDOT Geospatial Open Data Portal**:

- Highway data: ”WSDOT - State Route Linear Referencing System (LRS) Current”
- Ramp data: ”WSDOT - State Route Ramp (1:24K) Current”

Highway Sampling

Approach:

1. Load geospatial data for both sides of Highway 14
2. Filter these into the 5 areas using the milepost boundaries for what ECCA cleans
3. Divide each area into 600 foot segments
4. Randomly select one of these segments in each area

Highway Sampling: Segments

area	area ID	start MP	end MP	total miles	total segments
Camas (north)	H1	10	11.99	1.99	17
Camas (south)	H2	10.09	10.58	0.49	4
Washougal “extra” (south)	H3	13.75	14.3	0.55	4
Washougal (north)	H4	15.05	17	1.95	17
Washgoual (south)	H5	15.05	17	1.95	17

Ramp Sampling

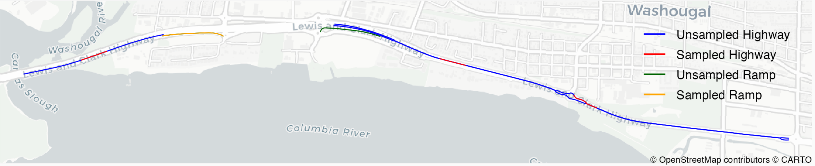
Idea

Pick one ramp from each pair, but make sure at least one on-ramp and one off-ramp are selected

Sampling Results: West Side

- 2 highway samples
- 2 ramp samples

Sampling Results: East Side



- 3 highway samples
- 1 ramp sample

Sampling: Further Details

In our final report you can find:

- Link to the code used to generate the samples
- A more technical description of the statistical methodology